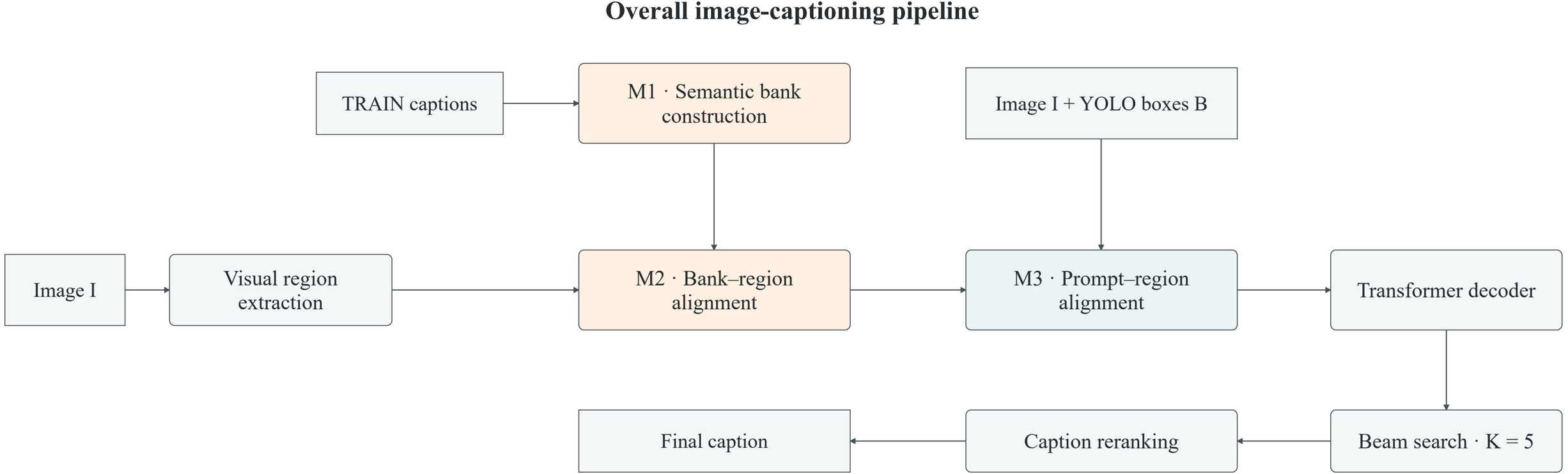

Overall image-captioning pipeline
TRAIN captions
M1 · Semantic bank construction
Image I + YOLO boxes B
Image I
Visual region extraction
M2 · Bank–region alignment
M3 · Prompt–region alignment
Transformer decoder
Final caption
Caption reranking
Beam search · K = 5